

COMPUTER NETWORKS LABORATORY

CN LAB – EXPERIMENT 15

Operating System Detection using Nmap

A concise practical record for GitHub and LMS submission

Experiment	15
Title	Operating System Detection using Nmap
Tool	Nmap (Network Mapper)
Category	Network Security / Reconnaissance
Submission	GitHub / LMS

Note: Run Nmap only against systems you own or have explicit permission to test.

1. Aim

To detect and identify the probable operating system of an authorized target host using Nmap OS detection techniques.

2. Objectives

- Verify that Nmap is installed and available from the terminal.
- Identify an authorized target host for the laboratory exercise.
- Perform operating-system detection using Nmap.
- Interpret the important fields reported in the scan output.

3. Requirements

Requirement	Purpose
Nmap	Network scanning and OS fingerprinting
Windows / Linux / macOS	Operating environment
Terminal / Command Prompt	Run Nmap commands
Authorized target	Host to be tested

4. Nmap Installation / Verification

Open a terminal or Command Prompt and verify the installation:

```
nmap --version
```

A successful installation displays the installed Nmap version and related information.

5. Basic Host Check

Replace with the IP address or hostname of the authorized laboratory target.

```
nmap -sn <TARGET_IP>
```

-sn performs host discovery without a port scan.

6. Operating System Detection

Step 1: Open Command Prompt / Terminal.

Step 2: Select an authorized target system.

Step 3: Run the OS detection command.

Step 4: Review the detected OS family, device type and confidence information.

Primary command:

```
nmap -O <TARGET_IP>
```

The **-O** option enables Nmap operating-system detection. The result is an estimate based on TCP/IP fingerprinting and may vary with network conditions and filtering.

7. Optional Detailed Scan

For an authorized lab target, OS detection can be combined with service/version detection:

```
nmap -O -sV <TARGET_IP>
```

-sV attempts to identify services and their versions in addition to OS detection.

8. Sample Output

```
Starting Nmap 7.x
Nmap scan report for <TARGET_IP>
Host is up.

PORT      STATE  SERVICE
22/tcp    open   ssh
80/tcp    open   http

Device type: general purpose
Running: Linux
OS CPE: cpe:/o:linux:linux_kernel
OS details: Linux-based operating system
Network Distance: 1 hop

OS detection performed.
```

Note: This is sample output. Actual results depend on the authorized target, network, firewall and Nmap fingerprint database.

9. Output Explanation

Field	Meaning
Host is up	The target responded to host discovery / scanning.
PORT / STATE / SERVICE	Shows discovered ports, their state and associated service.
Device type	Nmap's estimated category of the target device.
Running	Probable operating-system family detected by fingerprinting.
OS details	More specific OS information when available.
Network Distance	Estimated number of network hops to the target.

10. Result

The operating system of an authorized target host was analyzed using Nmap OS detection. Nmap uses network responses and fingerprint matching to estimate the most likely operating-system family and device characteristics.

11. Precautions

- Scan only systems for which you have authorization.
- Use the correct target IP address or hostname.

- Do not scan public or third-party systems without permission.
- OS detection is an estimate and can be affected by firewalls and network conditions.